AWAD WAGDY AWAD

Full-Stack Software Engineer & Desktop Systems Specialist (Offline & Online) | Computer Vision & LiDAR QA Expert

Email: twagdy067@gmail.com | Phone: +20 120 233 2355 | Location: Ain Shams, Cairo, Egypt | Web: awad-wagdy-portfolio.vercel.app

PROFESSIONAL SUMMARY

Software Engineer and Quality Assurance professional specialized in two high-impact tracks: Track 1 - Custom Desktop Systems & Full-Stack Web: Architecting and delivering robust standalone desktop applications (both 100% offline-first with zero-latency embedded SQLite databases and cloud-synchronized online systems) tailored for enterprise and commercial businesses (e.g., AL FARES ERP & POS Suite), alongside high-converting commercial web platforms and digital interactive flipbooks (e.g., Al-Amin Pergolas official platform and catalog). Track 2 - Computer Vision, LiDAR Point Clouds & Perception QA: Strengthening autonomous vehicle deep learning datasets through rigorous 2D/3D cuboid audits, LiDAR 3D point cloud labeling, and dense semantic segmentation validation at Micro Engineering Egypt.

WORK EXPERIENCE

Quality Control Technician | MICRO ENGINEERING EGYPT **10/2025 - Present**

- Checked annotation outputs for consistency with task instructions before final approval.
- Identified precision labeling errors and flagged ambiguous corner cases for engineering rework.
- Compared multi-sensor outputs against raw LiDAR 3D point clouds to verify spatial accuracy.
- Documented recurring quality error modes and shared benchmark reports with the team.

Data Labeling Quality Assurance Specialist | MICRO ENGINEERING EGYPT **2024 - 09/2025**

- Audited and refined 2D/3D bounding boxes, LiDAR point cloud tracks, and semantic segmentation masks.
- Enforced quality inspection gates to maintain 99%+ accuracy across project batches.
- Validated 2D and 3D bounding box orientations, yaw angles, and bounding contours for vehicles and pedestrians.

Data Labeling Specialist | MICRO ENGINEERING EGYPT **2023 - 2024**

- Annotated visual and sensor data for autonomous driving deep learning models.
- Executed scene classification, road free-space segmentation, and static/dynamic object tagging.

KEY DELIVERED SOFTWARE SYSTEMS

AL FARES ERP & POS Integrated Desktop Suite:
Complete commercial desktop application built with 10 production modules: POS barcode cashier terminal, live profit & loss analytics, warehouse inventory balance audit, supplier purchases, customer debt accounts with credit limits, daily treasury disbursements, and automatic PDF receipt archiving. 100% offline-first.

Al-Amin Pergolas Commercial Web Platform:
Bespoke architectural showcase website featuring high-performance architectural galleries, responsive modern layout, and direct WhatsApp quote generation.

Al-Amin Interactive Digital Flipbook Catalog:
Interactive digital flipbook catalog application with realistic 3D book turning, wood frame design portfolio, lightbox zoom, and instant quote inquiry.

EDUCATION & CREDENTIALS

Bachelor of Management Information Systems (MIS) | Taibah Integrated Science Academy **Jan 2022**

CORE TECHNICAL SKILLS & TOOLS

Desktop & Web Development: Custom Desktop Apps (Offline & Online), SQLite Local Database, Role-Based Access Control (RBAC), POS Thermal ESC/POS Printers, Barcode Scanners, React, TypeScript, Tailwind CSS, REST APIs.

Computer Vision & QA: 3D LiDAR Point Clouds, 2D/3D Bounding Boxes, Semantic Segmentation, Sensor Fusion, Error Taxonomy, Dataset Benchmarking.

Platforms & Tools: CVAT, Labelbox, Scale AI, Apollo Auto, Git, GitHub, JIRA, Python Scripting.